

the receiver clock to synchronise with the phase of incoming data. The differential sign of D+/D- signal is then sampled according to the sync pattern. HSIC uses a separate strobe line to tell the receiver when to sample incoming data. HSIC data signal is sampled at the rising and falling edges of the strobe signal. If the strobe and data signals become skewed for any reason, sampled data may become corrupted. HSIC Electrical Specification defines the maximum allowable skew as 15ps.

To make sure the skew does not become an issue, HSIC traces must be kept as short as possible and must not be longer than 10cm. Data and strobe traces must be the same length, and these should be routed to 50 Ω single-ended impedance.

To illustrate the amount of skew possible in the real world, Fig. 2 shows the beginning of a test packet transmitted from a host to a device with equal lengths.

The same packet transmitted from the same host with a strobe trace that is about 10cm longer than the data trace is shown in Fig. 3. The resulting skew is about half of a nanosecond. This is an extreme example, but results suggest that even a small amount of length mismatch may result in an HSIC specification violation.

The single-ended nature and differences in signal termination cause some difficulties when attempting to probe HSIC lines. Standard USB signals can be easily monitored and

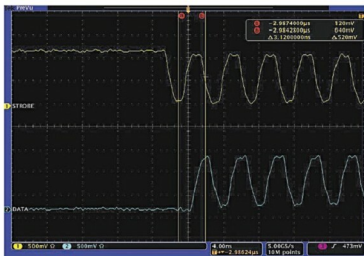


Fig. 2: Equal HSIC trace lengths

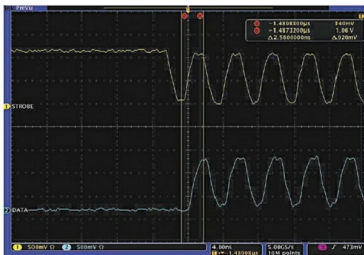


Fig. 3: Strobe trace 10cm longer than data trace

deciphered by placing a differential probe connected to an oscilloscope at either the transmitter side or the receiver side. HSIC signals are more sensitive and, thus, transmission line theory should be considered when attempting to probe these.

A good general guideline is to probe at the side opposite to the source of the signal that needs to be observed. For instance, to observe the signals originating from a device, place a probe at host-side terminals. To observe the signals originating from a host, place the probes at device-side terminals.

When attempting to probe signals originating from a device

while probing at the device side, the signal becomes distorted. This is likely due to the interference caused by the signal reflecting back on itself. The middle of the trace can also be probed, but results are typically not as clean as if probed properly from one side. The ideal would be to probe simultaneously from both ends. A series protocol analyser may be able to sample the signals accurately in both directions, but the 10cm trace length restriction makes this option impractical.

Making the connection

HSIC interface is structured such that a host or peripheral can be powered on in any order. To ensure a false connection is not detected, the host, hub and peripherals must ensure that the strobe or data lines do not float to an undeter-

mined value, commonly referred to as tri-stated.

Fig. 4 shows an oscilloscope capture of a connect sequence. This connect sequence is much simpler than the USB connect sequence because there are no speeds to negotiate. This sequence can be handled by a very simple state machine, reducing die size requirements.

With standard USB, the host can determine if a downstream port has been disconnected by monitoring the magnitudes of DP/DM signal voltages. If the voltage exceeds the disconnect voltage threshold, the host can conclude that the device has been disconnected.